

Bernoulli PMF $X \sim \text{Bern}(p)$	Bernoulli Mean $X \sim \text{Bern}(p)$
Bernoulli Variance $X \sim \text{Bern}(p)$	Beta PDF $X \sim \text{Beta}(\alpha, \beta)$
Beta Mean $X \sim \text{Beta}(\alpha, \beta)$	Beta Variance $X \sim \text{Beta}(\alpha, \beta)$
Beta MGF $X \sim \text{Beta}(\alpha, \beta)$	Bernoulli MGF $X \sim \text{Bern}(p)$
Binomial PMF $Y \sim \text{Bin}(n, p)$	Binomial Mean $X \sim \text{Bin}(n, p)$

$\mathbb{E}[X] = p$	$f_X(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$ $\Omega = \{p : 0 < p < 1\}$
$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$ $\Omega = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$	$\text{Var}(X) = p(1-p)$
$\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}$
$M_X(t) = 1 - p + pe^t$	No nice closed form MGF
$\mathbb{E}[Y] = np$	$f_Y(y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, \dots, n$ $\Omega = \{(n, p) : n \in \mathbb{Z}^+, 0 < p < 1\}$

Binomial Variance $X \sim \text{Bin}(n, p)$	Binomial MGF $X \sim \text{Bin}(n, p)$
Cauchy PDF $X \sim \text{Cauchy}(\theta, \sigma)$	Cauchy Mean $X \sim \text{Cauchy}(\theta, \sigma)$
Cauchy Variance $X \sim \text{Cauchy}(\theta, \sigma)$	Cauchy MGF $X \sim \text{Cauchy}(\theta, \sigma)$
χ^2 PDF $X \sim \chi^2_\nu$	χ^2 Mean $X \sim \chi^2_\nu$
χ^2 Variance $X \sim \chi^2_\nu$	χ^2 MGF $X \sim \chi^2_\nu$

$M_Y(t) = (1 - p + pe^t)^n$	$\text{Var}(Y) = np(1 - p)$
DNE	$f_X(x) = (\pi\sigma)^{-1} \left(1 + \left(\frac{x-\theta}{\sigma}\right)^2\right)^{-1}, \quad x \in \mathbb{R}$ $\Omega = \{(\theta, \sigma) : \theta \in \mathbb{R}, \sigma > 0\}$
No nice closed form MGF	DNE
$\mathbb{E}[X] = \nu$	$f_X(x) = (\Gamma(\nu/2) 2^{\nu/2})^{-1} x^{(\nu/2)-1} e^{-x/2}, \quad x > 0$ $\Omega = \{\nu : \nu \in \mathbb{Z}^+\}$
$M_X(t) = \left(\frac{1}{1-2t}\right)^{\nu/2}, \quad t < 1/2$	$\text{Var}(X) = 2\nu$

Discrete Uniform PMF $U \sim \text{dUnif}(a, b)$	Discrete Uniform Mean $U \sim \text{dUnif}(a, b)$
Discrete Uniform Variance $U \sim \text{dUnif}(a, b)$	Discrete Uniform MGF $U \sim \text{dUnif}(a, b)$
Exponential PDF $X \sim \text{Exp}(\theta)$	Exponential Mean $X \sim \text{Exp}(\theta)$
Exponential Variance $X \sim \text{Exp}(\theta)$	Exponential MGF $X \sim \text{Exp}(\theta)$
Snedecor's F PDF $X \sim F(\nu_1, \nu_2)$	Snedecor's F Mean $X \sim F(\nu_1, \nu_2)$

$\mathbb{E}\left[U\right]=\frac{a+b}{2}$	$f_U(u)=(b-a+1)^{-1},\quad u=a,a+1,\cdots,b$ $\Omega=\{(a,b):a,b\in\mathbb{Z},a<b\}$
$M_U(t)=\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)},\quad t<0$	$\text{Var}\left(U\right)=\frac{(b-a+1)^2-1}{12}$
$\mathbb{E}\left[X\right]=\theta$	$f_X(x)=\theta^{-1}e^{-x/\theta},\quad 0\leq x<\infty$ $\Omega=\{\theta:\theta>0\}$
$M_X(t)=\frac{1}{1-\theta t},\quad t<1/\theta$	$\text{Var}\left(X\right)=\theta^2$
$\mathbb{E}\left[X\right]=\frac{\nu_2}{\nu_2-2},\nu_2>2$	$f_X(x)=\frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)}{\Gamma(\nu_1/2)\Gamma(\nu_2/2)}\left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2}\frac{x^{(\nu_1-2)/2}}{\left(1+\left(\frac{\nu_1}{\nu_2}\right)x\right)^{(\nu_1+\nu_2)/2}},$ $x\in\mathbb{R}^+$ $\Omega=\{(\nu_1,\nu_2):\nu_1,\nu_2\in\mathbb{Z}^+\}$

Snedecor's F Variance $X \sim F(\nu_1, \nu_2)$	Snedecor's F MGF $X \sim F(\nu_1, \nu_2)$
What are the parameters of the Gamma distribution	Gamma PDF $X \sim \text{Unif}(\alpha, \theta)$
Gamma Mean $X \sim \text{Unif}(\alpha, \theta)$	Gamma Variance $X \sim \text{Unif}(\alpha, \theta)$
Gamma MGF $X \sim \text{Unif}(\alpha, \theta)$	Gamma distribution relationship with other distributions
What is p in the Geometric Distribution	Geometric PMF $X \sim \text{Geom}(p)$

No nice closed form MGF	$\text{Var}(X) = 2 \left(\frac{\nu_2}{\nu_2 - 2} \right)^2 \frac{\nu_1 + \nu_2 - 2}{\nu_1(\nu_2 - 4)}, \quad \nu_2 > 4$
$f_X(x) = (\Gamma(\alpha) \theta^\alpha)^{-1} x^{\alpha-1} e^{-x/\theta}, \quad x > 0$ $\Omega = \{(\alpha, \theta) : \alpha > 0, \theta > 0\}$	The parameterization I will use is • α is the shape parameter • θ is the scale parameter • Note: β is the rate parameter and $\beta = 1/\theta$
$\text{Var}(X) = \alpha\theta^2$	$\mathbb{E}[X] = \alpha\theta$
1. If $\alpha = 1$, then $X \sim \text{Exp}(\theta)$ 2. If $X \sim \text{Gamma}(p/2, 2)$, then $X \sim \chi_p^2$ 3. If $X \sim \text{Gamma}(\alpha, \theta)$ where α is an integer, then $P(X \leq x) = P(Y \geq \alpha)$ where $Y \sim \text{Pois}(x/\theta)$	$M_X(t) = \left(\frac{1}{1 - \beta t} \right)^\alpha, \quad t < 1/\beta$
$f_X(x) = p(1-p)^{x-1}, \quad x = 1, 2, \dots$ $\Omega = \{p : 0 < p < 1\}$	The probability that it takes exactly x trials to obtain the first success

Geometric Mean $X \sim \text{Geom}(p)$	Geometric Variance $X \sim \text{Geom}(p)$
Geometric MGF $X \sim \text{Geom}(p)$	What do the Hypergeometric parameters and random variable represent?
Hypergeometric PMF $X \sim \text{HG}(N, K, n)$	Hypergeometric Mean $X \sim \text{HG}(N, K, n)$
Hypergeometric Variance $X \sim \text{HG}(N, K, n)$	Hypergeometric MGF $X \sim \text{HG}(N, K, n)$
Laplace (Double Exponential) PDF $X \sim \text{Lap}(\mu, \sigma)$	Laplace (Double Exponential) Mean $X \sim \text{Lap}(\mu, \sigma)$

$\text{Var}(X) = \frac{1-p}{p^2}$	$\mathbb{E}[X] = 1/p$
• N = total population • n = sample size • K = total number of successes in population • X = number of successes in sample	$M_X(t) = \frac{pe^t}{1-(1-p)e^t}, \quad t < -\log(1-p)$
$\mathbb{E}[X] = \frac{nK}{N}$	$f_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \dots, n$ $\Omega = \{(N, K, n) : K - (N - n) \leq x \leq K; N, K, n \geq 0\}$
No nice closed form MGF	$\text{Var}(X) = \frac{nK}{N} \frac{(N-K)(N-n)}{N(N-1)}$
$\mathbb{E}[X] = \mu$	$f_X(x) = (2\sigma)^{-1} \exp\{- x - \mu /\sigma\}, \quad x \in \mathbb{R}$ $\Omega = \{(\mu, \sigma) : \mu \in \mathbb{R}, \sigma > 0\}$

Laplace (Double Exponential) Variance $X \sim \text{Lap}(\mu, \sigma)$	Laplace (Double Exponential) MGF $X \sim \text{Lap}(\mu, \sigma)$
Logistic PDF $X \sim \text{Log}(\mu, \beta)$	Logistic Mean $X \sim \text{Log}(\mu, \beta)$
Logistic Variance $X \sim \text{Log}(\mu, \beta)$	Logistic MGF $X \sim \text{Log}(\mu, \beta)$
Lognormal PDF $X \sim \text{LogN}(\mu, \sigma^2)$	Lognormal Mean $X \sim \text{LogN}(\mu, \sigma^2)$
Lognormal Variance $X \sim \text{LogN}(\mu, \sigma^2)$	Lognormal MGF $X \sim \text{LogN}(\mu, \sigma^2)$

$M_X(t) = \frac{e^{\mu t}}{1-(\sigma t)^2}, \quad t < \sigma^{-1}$	$\text{Var}(X) = 2\sigma^2$
$\mathbb{E}[X] = \mu$	$f_X(x) = \beta^{-1} \frac{\exp\{-(x-\mu)/\beta\}}{(1+\exp\{-(x-\mu)/\beta\})^2}, \quad x \in \mathbb{R}$ $\Omega = \{(\mu, \beta) : \mu \in \mathbb{R}, \beta > 0\}$
$M_X(t) = e^{\mu t} \Gamma(1 - \beta t) \Gamma(1 + \beta t), \quad t < \beta^{-1}$	$\text{Var}(X) = \frac{\pi^2 \beta^2}{3}$
$\mathbb{E}[X] = \exp\{\mu + (\sigma^2/2)\}$	$f_X(x) = (2\pi\sigma^2)^{-1/2} \frac{\exp\{-(\log(x-\mu))^2/2\sigma^2\}}{x}, \quad x \in \mathbb{R}^+$ $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$
No nice closed form MGF	$\text{Var}(X) = \exp\{2(\mu + \sigma^2)\} - \exp\{2\mu + \sigma^2\}$

The random variable of the Negative Binomial Distribution	Negative Binomial PMF $X \sim \text{NB}(r, p)$
Negative Binomial Mean $X \sim \text{NB}(r, p)$	Negative Binomial Variance $X \sim \text{NB}(r, p)$
Negative Binomial MGF $X \sim \text{NB}(r, p)$	Normal PDF $X \sim N(\mu, \sigma^2)$
Normal Mean $X \sim N(\mu, \sigma^2)$	Normal Variance $X \sim N(\mu, \sigma^2)$
Normal MGF $X \sim N(\mu, \sigma^2)$	Standard Normal Random Variable Z

$f_X(x) = \binom{r+x-1}{x} p^r (1-p)^x, \quad x = 0, 1, \dots$ $\Omega = \{(r, p) : r \in \mathbb{Z}^+, r \geq 1, 0 < p < 1\}$	• X = the number of failures to get the r^{th} success. • Y = the number of trials to get r^{th} success. • Note that $Y = X + r$ since r is the number of successes
$\text{Var}(X) = \frac{r(1-p)}{p^2}$	$\mathbb{E}[X] = \frac{r(1-p)}{p}$
$f_X(x) = (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}, \quad x \in \mathbb{R}$ $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$	$M_X(t) = \left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < -\log(1-p)$
$\text{Var}(X) = \sigma^2$	$\mathbb{E}[X] = \mu$
If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$	$M_X(t) = \exp\left\{\mu t + \frac{\sigma^2 t^2}{2}\right\}$

Pareto PDF $X \sim \text{Par}(\alpha, \beta)$	Pareto Mean $X \sim \text{Par}(\alpha, \beta)$
Pareto Variance $X \sim \text{Par}(\alpha, \beta)$	Pareto MGF $X \sim \text{Par}(\alpha, \beta)$
What do the parameter and random variable of the Poisson distribution represent?	Poisson PMF $X \sim \text{Pois}(\lambda)$
Poisson Mean $X \sim \text{Pois}(\lambda)$	Poisson Variance $X \sim \text{Pois}(\lambda)$
Poisson MGF $X \sim \text{Pois}(\lambda)$	Student's T PDF $X \sim t(\nu)$

$\mathbb{E}[X] = \frac{\beta\alpha}{\beta-1}, \quad \beta > 1$	$f_X(x) = x^{-(\beta+1)}\beta\alpha^\beta, \quad x > \alpha$ $\Omega = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$
No nice closed form MGF	$\text{Var}(X) = \frac{\beta\alpha^2}{(\beta-1)^2(\beta-2)}, \quad \beta > 2$
$f_X(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, \dots$ $\Omega = \{\lambda : \lambda > 0\}$	• X is the number of occurrences of an event in a given interval of time • λ is the expected number of an event (usually rare) in a given interval of time
$\text{Var}(X) = \lambda$	$\mathbb{E}[X] = \lambda$
$f_X(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\nu/2)} (\nu\pi)^{-1/2} (1+x^2/\nu)^{-(\nu+1)/2}, \quad x \in \mathbb{R}$ $\Omega = \{\nu : \nu \in \mathbb{Z}^+\}$	$M_X(t) = \exp\{\lambda(e^t - 1)\}$

Student's T Mean $X \sim t(\nu)$	Student's T Variance $X \sim t(\nu)$
Student's T MGF $X \sim t(\nu)$	Uniform PDF $U \sim \text{Unif}(a, b)$
Uniform Mean $U \sim \text{Unif}(a, b)$	Uniform Variance $U \sim \text{Unif}(a, b)$
Uniform MGF $U \sim \text{Unif}(a, b)$	Weibull PDF $X \sim \text{Weib}(\gamma, \beta)$
Weibull Mean $X \sim \text{Weib}(\gamma, \beta)$	Weibull Variance $X \sim \text{Weib}(\gamma, \beta)$

$\text{Var} (X) = \frac{\nu}{\nu-2}, \quad \nu > 2$	$\mathbb{E} [X] = 0, \quad \nu > 1$
$f_U(u) = (b-a)^{-1}, \quad u \in [a,b]$ $\Omega = \{(a,b) : a,b \in \mathbb{R}, a < b\}$	No nice closed form MGF
$\text{Var} (U) = \frac{(b-a)^2}{12}$	$\mathbb{E} [U] = \frac{a+b}{2}$
$f_X(x) = \gamma \beta^{-1} x^{\gamma-1} \exp(-x^\gamma/\beta), \quad x \geq 0$ $\Omega = \{(\gamma, \beta) : \gamma > 0, \beta > 0\}$	$M_U(t) = \frac{e^{bt}-e^{at}}{(b-a)t}, \quad t \neq 0$
$\text{Var} (X) = \beta^{2/\gamma} [\Gamma(1+2\gamma^{-1}) - \Gamma^2(1+\gamma^{-1})]$	$\mathbb{E} [X] = \beta^{1/\gamma} \Gamma(1+\gamma^{-1})$

Weibull MGF $X \sim \text{Weib}(\gamma, \beta)$	

	No nice closed form MGF